

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	29 September 2026
Team ID	SWTID-2026-4375
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

Technical Architecture

Technical Architecture - Client-side validation on the ServiceNow Incident form

Implement Client Script & UI Policy (Incident)

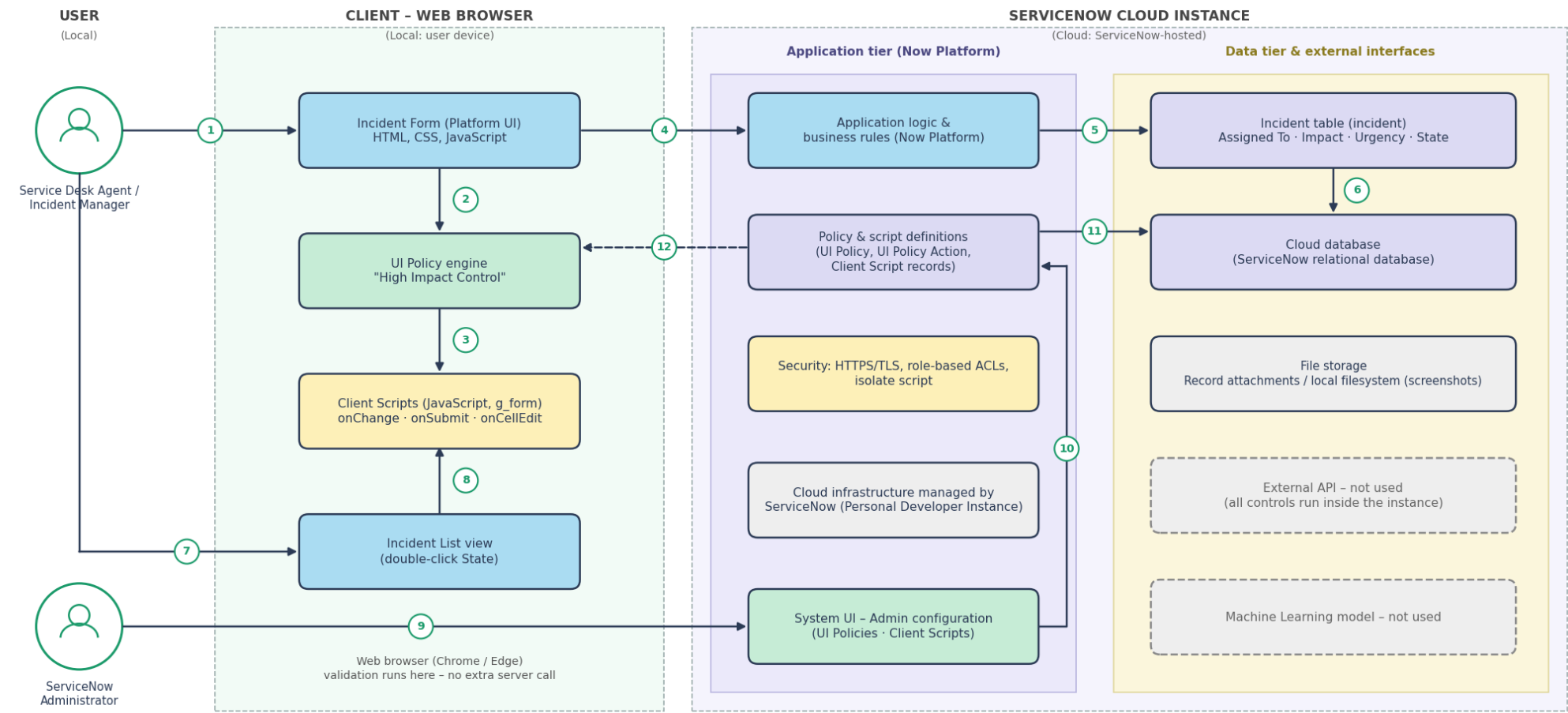


Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	Incident form in ServiceNow used by the Service Desk Agent / Administrator through a web browser	ServiceNow Platform UI (HTML, CSS, JavaScript)
2.	Application Logic-1	UI Policy "High Impact Control": when Impact = 1 - High, Assignment group becomes mandatory and Urgency becomes read-only (Reverse if false)	ServiceNow UI Policy & UI Policy Actions
3.	Application Logic-2	onChange Client Script on Impact: auto-sets Urgency to High and shows an info message	Client Script (JavaScript, g_form API)
4.	Application Logic-3	onSubmit script blocks save if Assigned To is empty for High impact; onCellEdit script blocks State edits in list view	Client Script - onSubmit & onCellEdit (JavaScript)
5.	Database	Incident table stores records; policy and script definitions are stored in platform tables	ServiceNow relational database (Now Platform)
6.	Cloud Database	Database service hosted with the ServiceNow cloud instance	ServiceNow Cloud (Personal Developer Instance)
7.	File Storage	No special file storage needed; screenshots and documents are kept locally or as record attachments	Local filesystem / ServiceNow attachments
8.	External API-1	Not used - all controls run within the ServiceNow instance	Not applicable
9.	External API-2	Not used - no external integration required	Not applicable
10.	Machine Learning Model	Not used in this project	Not applicable
11.	Infrastructure (Server / Cloud)	Deployed on a ServiceNow-hosted cloud instance. Local server configuration: not required. Cloud server configuration: managed by ServiceNow	ServiceNow Cloud instance

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No external open-source framework; uses JavaScript with the ServiceNow client-side APIs (g_form)	JavaScript (ServiceNow client scripting)
2.	Security Implementations	Role-based access so only administrators can configure policies and scripts; scripts run in isolated mode; HTTPS access to the instance	Role-Based Access Control (ACLs), Isolate script, HTTPS
3.	Scalable Architecture	Multi-instance cloud platform; validation runs in the user's browser, reducing server load, and applies to every Incident record	ServiceNow Now Platform (cloud, multitier)
4.	Availability	Instance is hosted and maintained by ServiceNow with redundant infrastructure; no extra services to depend on	ServiceNow cloud hosting
5.	Performance	Lightweight client-side rules respond instantly on the form with no extra server round-trip for validation	UI Policy, Client Scripts